

Retail Customer & Sales Intelligence

Combined Project Brief — Revised Edition

Industry-Level Data Analyst Project

Retail Customer & Sales Intelligence using the Uploaded Retail Dataset

Case study: Analyze retail purchasing behavior and identify the products, customers, locations and business opportunities that can help management improve sales and customer value.

Dataset	Records	Columns	Domain
Retail customer purchases	3,900	18	Customer & Sales Analytics

Business Case & Problem Statement

You are working as a Data Analyst for a retail company. Management wants to understand customer purchasing behavior, product performance, customer segments, promotions, locations, and operational preferences. The leadership team needs evidence-based answers that can support sales growth, customer targeting, product decisions and future investment.

Your objective: Find the important patterns hidden in the data, quantify them, identify business opportunities and problems, and translate the findings into practical recommendations.

Project Overview

- **Business goal:** Understand what drives purchase value and where the company has growth opportunities.
- **Primary focus:** Sales performance, customer segments, products/categories, promotions, locations, seasonality and purchasing behavior.
- **Expected outcome:** A management-ready set of insights explaining what is happening, where the opportunities are, and what should be investigated or improved.
- **Analytical principle:** Every important conclusion should be supported by measurable evidence from the dataset.

Dataset Profile

Data area	Available information
Customer	Customer ID, Age, Gender, Subscription Status
Product	Item Purchased, Category, Size, Color
Purchase	Purchase Amount (USD), Previous Purchases, Frequency of Purchases
Location & season	Location, Season
Promotion	Discount Applied, Promo Code Used
Operations	Shipping Type, Payment Method
Customer feedback	Review Rating

Important Dataset Limitation

The current dataset does not contain transaction dates, order IDs, product costs, profit/margin, returns/refunds or detailed marketing spend. Therefore, true profit, ROI, month-over-month growth, customer retention and lifetime value cannot be directly established from this dataset alone.

Core Business Questions You Should Answer

1. Overall Business Performance

1. What is the total sales / total purchase value represented in the dataset?

2. What is the average purchase value per transaction/customer record?
3. Which categories, products and locations contribute the largest share of total sales?
4. How concentrated are sales? Does a small number of products or categories generate a large portion of total sales?
5. Which parts of the business appear to be the strongest contributors to overall sales?

2. Customer Segments & Investment

1. Which customer segment spends the most?
2. Which age group has the highest average purchase value and which has the highest total purchase value?
3. Do purchasing patterns differ between male and female customers?
4. Do subscribers behave differently from non-subscribers in purchase value and previous purchases?
5. Which customer segment should the company prioritize for additional investment, and what evidence supports that decision?
6. Which customer segment appears underdeveloped but has potential for growth?

3. How Can the Company Increase Sales?

1. Which products or categories have the greatest opportunity to drive additional sales?
2. Which customer segments should be targeted to increase purchase value?
3. Are there categories that perform strongly with one customer segment but weakly with another?
4. Which locations have high customer volume but relatively low average purchase value?
5. Which locations have high average purchase value but relatively low customer volume?
6. What are the 3 most realistic actions the company could investigate to increase sales based on the data?

4. Product & Category Performance

1. Which product is the highest contributor to total sales?
2. Which categories have the highest average purchase value?
3. Which products combine high sales with high customer ratings?
4. Which sizes and colors are most frequently purchased?

5. Discounts & Promotions

1. Do customers using discounts have a different average purchase value from customers who do not?
2. Do customers using promo codes show different purchasing behavior?
3. Which customer segments are most associated with discount or promo-code usage?

6. Customer Loyalty & Purchasing Behavior — Power BI Visualisation Questions

1. Which Power BI visual can show how average and total purchase value vary across previous-purchase groups?
2. How can a bar or column chart compare average and total purchase value across purchase-frequency groups?
3. Which visual can display the relationship between previous purchases and current purchase value?
4. How can a matrix or stacked bar chart compare customer groups by current purchase value and purchase history?
5. Which combination of cards, charts and slicers can profile the characteristics of higher-value customers?

7. Location & Market Opportunities — Power BI Visualisation Questions

1. Which map visual can compare total purchase value across locations?
2. How can a bar chart rank locations by average purchase value?
3. Which scatter plot can highlight locations with high customer volume but low average purchase value?
4. How can a scatter plot identify locations with low customer volume but high average purchase value?
5. How can a map or scatter plot combine customer volume and purchase behavior to explore potential market-expansion opportunities?

8. Seasonality & Operations — Power BI Visualisation Questions

1. Which column chart can compare total and average purchase value across seasons?
2. How can a clustered column chart or heatmap compare category performance across seasons?
3. Which bar chart can show the most-used shipping methods and compare purchase value by shipping type?
4. How can a bar chart compare payment methods among higher-value purchases?
5. Which Power BI visuals can bring together shipping, payment and purchase-value patterns for operational exploration?

9. Final Management Questions

1. If management gives you a fixed budget for growth, where would you recommend investigating investment first, and why?
2. What are the top 5 insights management should know from this dataset?
3. What are the top 3 actions that could potentially improve sales or customer value?
4. What products, customer groups or locations should management monitor closely?
5. What important business questions cannot be answered reliably from the current data, and what additional information would be needed?

Insights You Should Specifically Look For

- **Sales concentration:** Determine whether a small number of categories, products or locations account for a large share of total purchase value.
- **High-value customer segments:** Identify age, gender, subscription and purchase-frequency groups with materially different average or total purchase values.
- **Investment opportunities:** Look for customer/location segments where the combination of volume and purchase value suggests room for growth.
- **Sales growth opportunities:** Identify products, categories and customer groups where targeted actions could potentially increase purchase value.
- **Promotion behavior:** Compare discount and promo-code users with non-users and determine where the differences are largest.
- **Customer loyalty pattern:** Investigate whether previous purchase history is associated with higher current purchase value.
- **Product quality signal:** Find products/categories with high sales but weaker ratings that may require investigation.
- **Location opportunity:** Separate high-volume/low-value markets from low-volume/high-value markets.
- **Seasonal patterns:** Identify categories or products that perform differently across seasons.
- **Operational patterns:** Understand dominant shipping and payment methods and their relationship with purchase value.
- **Actionable insight:** Observation → Evidence → Business implication → Recommended action.

Recommended Customer Segmentation

Create customer segments that help management compare behavior and identify where attention or investment could be focused. Use the dataset's actual values to define the final thresholds rather than assuming that one fixed threshold is automatically correct.

Segmentation dimension	Suggested groups	What to investigate
Age	18–24, 25–34, 35–44, 45–54, 55+	Purchase value, category preference, frequency
Purchase value	Low / Medium / High	Characteristics of higher-value customers
Previous purchases	Low / Medium / High	Relationship with current purchase value
Purchase frequency	Use the dataset's frequency groups	Recurring vs less-frequent behavior
Subscription	Subscriber / Non-subscriber	Purchase value and previous-purchase differences
Engagement combination	High-value + high-history / other combinations	Identify valuable customer profiles

Recommended Business Insight Format

Example: “Segment X represents __% of records but contributes __% of purchase value. Its average purchase is __ compared with __ for the overall dataset. This indicates a potentially valuable customer group to investigate for targeted growth initiatives.”

Final Objective

By the end of the project, management should be able to understand where sales are coming from, who the most valuable customers are, which products and markets have opportunities, what patterns surround promotions and loyalty, and what actions deserve further investigation to improve business performance.